UNIVERSITI TUNKU ABDUL RAHMAN

EXAM SPECIMEN PAPER

UCCC 3073 / UCCG 3073 DATA SCIENCE

[Suggested Answers]

BACHELOR OF COMPUTER SCIENCE (HONS)
BACHELOR OF SCIENCE (HONS) STATISTICAL COMPUTING AND OPERATIONS RESEARCH

Instruction to Candidates:

This question paper consists of **FIVE (5)** questions.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**. Each question carries 25 marks.

Should a candidate answers more than ONE (1) question in Section B, marks will only be awarded for the FIRST (1) question in that section in the order the candidate submits the answers.

Candidates are allowed to use any type of scientific calculator.

Answer questions only in the answer booklet provided.

SECTION A (Answer ALL questions)

Q1. (a) (6 marks)

First stage of data science methodology. The business sponsors who need the analytic solution play the most critical role in this stage by defining the problem, project objectives and solution requirements from a business perspective.

Reason 1

This first stage lays the foundation for a successful resolution of the business problem. It helps clarify the goal of the entity asking the question.

Reason 2

To help guarantee the project's success, the sponsors should be involved throughout the project to provide domain expertise, review intermediate findings and ensure the work remains on track to generate the intended solution.

(b) (6 marks)

1. Volume = data size is BIG, like millions of records, GB to TB of data. It implies enormous volumes of data and these data is generated by machines, networks and human interaction on systems like social media.
2. Velocity = data can easily generate via real-time, utilising current network technologies such as distributed systems, gfs and etc. It deals with the pace at which data flows in from sources like business processes, machines, networks and human interaction with things like social media sites, mobile devices, etc. The flow of data is massive and continuous.
3. Variety = data can be presented in various different forms, such as pictures, video clips, pdf, metadata, and more. It refers to the many sources and types of data both structured and unstructured.
4. Variability = data must have sufficient attributes to be understood and interpreted
5. Veracity = the degree of certainty of data. how do we know if the data that we analyse is from the reliable source and how do we know if it is not the artificial facts? It refers to the biases, noise and abnormality in data. Is the data that is being stored, and mined meaningful to the problem being analysed.
6. Visualisation = how can we present the data in the way that it is easily interpret by non-technical users?
7. Value = data must worth for value, otherwise, it is not worthy to perform any analytical exercise

(c) (5 marks)

Sepal length = ratio, continuous
Sepal width = ordinal, categorical
Petal length = ordinal, categorical
Petal width = ratio, continuous
Species = nominal, categorical

(d)

(i) (2 marks)

$$x' = \frac{x - min}{max - min}$$

x	x'	
200	(200 - 200) / (1000 - 200)	0
300	(300 - 200) / (1000 - 200)	0.125
400	(400 - 200) / (1000 - 200)	0.25
600	(600 - 200) / (1000 - 200)	0.5
800	(800 - 200) / (1000 - 200)	0.75
1000	(1000 - 200) / (1000 - 200)	1

(ii) (2 marks)

$$\bar{x} = \frac{200 + 300 + 400 + 600 + 800 + 1000}{6} = 550$$

$$\sigma_x = \sqrt{\frac{(200-550)^2 + (300-550)^2 + (400-550)^2 + (600-550)^2 + (800-550)^2 + (1000-550)^2}{6}}$$

$$= 281.37$$

x	x'	
200	(200 - 550) / 281.37	-1.244
300	(300 - 550) / 281.37	-0.889
400	(400 - 550) / 281.37	-0.533
600	(600 - 550) / 281.37	0.178
800	(800 - 550) / 281.37	0.889
1000	(1000 - 550) / 281.37	1.599

(e)

(i) (2 marks)

Sorted data: 10, 11, 15, 35, 50, 55, 85, 88, 92, 169, 204, 250
Bin 1 : 10, 11, 15, 35
Bin 2 : 50, 55, 85, 88
Bin 3 : 92, 169, 204, 250

(ii) (2 marks)

Sorted data: 10, 11, 15, 35, 50, 55, 85, 88, 92, 169, 204, 250
Bin width = (max – min)/ bin size = (250 – 10) / 3 = 80
Bin 1 (any values in-between 10 and 90): 10, 11, 15, 35, 50, 55, 85, 88
Bin 2 (any values in-between 90 and 170) : 92, 169
Bin 3 (any values in-between 170 and 250): 204, 250

[Total : 25 marks]

Q2. (a)

(i) (5 marks)

Probability & Frequency Table

	Price			Type			Review		Purchase?	
	Yes	No		Yes	No		Yes	No	Yes	No
	1.99	2.5	Camera	0	2	Average	1	1	2	4
	5.9	4.2	Game	2	1	Good	0	2		
		4.99	Productivity	0	1	Excellent	1	1		
		3.49								
$\bar{x}$	3.95	3.80	Camera	0	0.5	Average	0.5	0.25	0.33	0.67
σ	2.76	1.06	Game	1	0.25	Good	0	0.5		
			Productivity	0	0.25	Excellent	0.5	0.25		

(ii) (5 marks)

$$\text{pdf}, f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{\frac{-(x-\bar{x})^2}{2\sigma^2}}$$

$$f(\text{price} = 4.99|\text{yes}) = 0.135, \qquad f(\text{price} = 4.99|\text{no}) = 0.199$$

$$p(\text{yes}|E) = \frac{0.135 * 1 * 0.5 * 0.33}{E} = \frac{0.022}{E}$$

$$p(\text{no}|E) = \frac{0.199 * 0.25 * 0.25 * 0.67}{E} = \frac{0.0083}{E}$$

$$p(ID7 == '\text{yes}') = \frac{0.022}{0.022 + 0.0083} = 0.728$$

There is a probability of 72.8% that Mr. N will make the purchase.

(b)

(i) (8 marks)

$$\beta_1 = \frac{(4 * 4.9825) - (2.875 * 6.6)}{(4 * 2.1631) - (2.875)^2} = 2.4685$$

$$\beta_0 = 1.65 - (2.4685 * 0.71875) = -0.12423$$

Therefore, the regression line for this data set is

$$y = -0.12423 + 2.4685x$$

(ii) (2 marks)

$$y = -0.12423 + (2.4685 * 1.5) = 3.5785$$

(iii) (5 marks)

$$R^2 = \left[\frac{(4 * 4.9825) - (2.875 * 6.6)}{\sqrt{[(4 * 2.1631) - (2.875)^2] * [(4 * 6.6) - (11.48)^2]}}\right]^2 = 0.9989$$

[Total : 25 marks]

Q3. (a)

(i) (2 marks)

Association rules are a mining technique “use to discover interesting associations between attributes contained in a database”.
Main ways in measuring association rules are:
-confidence (conditional probability B is true when A is true in an A ➔B statement)
-support (“minimum percentage of instances in the database that contain all items listed in a given association rule”)

(ii) (6 marks)

Transaction ID	Items
1	Anvil, TNT
2	Boomerang, Carrots (Iron), TNT
3	Dehydrated boulders, Earthquake pills, TNT
4	Earthquake pills, TNT

C_1			L_1	
Itemset	Support		Itemset	Support
~~{Anvil}~~	~~1~~		{Earthquake pills}	2
~~{Boomerang}~~	~~1~~		{TNT}	4
~~{Carrots (Iron)}~~	~~1~~	→		
~~{Dehydrated boulders}~~	~~1~~			
{Earthquake pills}	2			
{TNT}	4			

C_2			L_2	
Itemset	Support		Itemset	Support
{Earthquake pills, TNT}	2		{Earthquake pills, TNT}	2

Therefore, the frequent itemsets are:
{Earthquake pills}, {TNT}, {Earthquake pills, TNT}

(iii) (2 marks)

TNT and earthquake pills.

(iv) (2 marks)

Support and confidence.

(v) (3 marks)

The power of association rules may be in the eye of the data scientist. Requirements for minimum coverage, support, and confidence can all are set based on the tolerance or desire of an individual data miner. Thus, the ability of the technique to shed light on a problem may indeed be at the hands of the experimenter. More so, even numerically valued rule measures may be of little value when applied to the practical world.

To know that there is an 80% chance that someone who buys beer on a Sunday might also buy potato chips may be trivial to a company or individual. It is the odd occurrences (i.e., buying beer and diapers) that are of great value.

(b)

(i) (2 marks)
Class X

(ii) (2 marks)
Class O

(iv) (6 marks)
Bias is an error derived from erroneous assumptions made in the classification and high bias can result the missing of relevant relations between features and target output. Bias is the difference between the true target values and the predicted values.
Variance is an error derived from fluctuations in the training set and high variance can result the random noise generated in the training data. variance is the expectation of the squared deviation of a random variable from its mean.
When k value increase, the bias in the training set is likely to increase as higher k values cover wider neighbours to be considered in classification. Consequently, it reduces variability when predicting a new example (low variance).
When k value decrease, the bias tends to be small. However, when it comes to predict a new example, it has higher chance to be an error which causes high variance.
The model becoming more complex when k increases as it includes more neighbours in the classification.

[Total : 25 marks]

SECTION B (Choose only ONE question)

Q4. (a)

(i) (3 marks)

Line (I) is more likely corresponds to the error from the validation set because validation set is unknown to classifier during training and it tends to have high variance if it is significantly different from the training set.

(ii) (2 marks)

The training should be stopped when the validation error hits the lowest to avoid the classifier being over-fitted.

(iii) (3 marks)

The current learning model suffers from over-fitting because memorise the patterns in the training set and consequently produces high error in the validation set. This is a high variance problem because classifier's performance is relying on the fluctuations in the training set.

(iv) (2 marks)

Validation error rate should be used when reporting the error rate of the trained learner because it is used to provide unbiased evaluation of the trained learner.

(b)

(i) (4 marks)

		Prediction	
		dog	cat
Target	dog (TP)	4	0
	cat (TN)	1	5

$$misclassification\ rate = \frac{FP + FN}{TP + TN + FP + FN} = \frac{1 + 0}{4 + 5 + 1 + 0} = 0.10$$

(ii) (6 marks)

$$precision = \frac{TP}{TP + FP} = \frac{4}{4 + 1} = 0.80$$

$$recall = \frac{TP}{TP + FN} = \frac{4}{4 + 0} = 1.00$$

$$F_1 measure = 2 * \frac{precision * recall}{precision + recall} = 2 * \frac{(0.80 * 1.00)}{0.80 + 1.00} = 0.89$$

Q4. (c) (5 marks)

$$accuracy = \frac{TP + TN}{P + N}$$
$$= \frac{TP}{P + N} + \frac{TN}{P + N}$$
$$= \frac{TP}{P + N} * \frac{P}{P} + \frac{TN}{P + N} * \frac{N}{N}$$
$$= sensitivity\ .\ \frac{P}{P + N} + specificity\ .\frac{N}{P + N}$$

[Total : 25 marks]

Q5.

(a)

(i) (2marks)

Baseline accuracy = (2200 + 4800) / 10000 = 0.7

(ii) (4 marks)

$$misclassification = 1 - \frac{1010 + 4800}{10000} = 0.419$$

$$sensitivity = \frac{4800}{2200 + 4800} = 0.6857$$

$$specificity = \frac{1010}{1010 + 1990} = 0.3367$$

$$F_1 = \frac{2 * 4800}{(2 * 4800) + 1990 + 2200} = 0.6962$$

(iii) (5 marks)

No, because

- The baseline accuracy is 0.7, however the model obtained only 0.581 accuracy indicated that the model is not doing well for this data.
- The model has sensitivity of 0.6857 indicating that it is able to correctly identify customers who did not respond to the marketing campaign.
- However, low specificity of 0.3367 showed that the model cannot detect customers who response positively to the campaign.
- Poor performance of the model may be due to the imbalance of records in the data. this is supported by low F1 score.

* The model has a sensitivity of 0.6857 indicating that it is able to correctly identify customers who respond positively to the marketing campaign. However, a high FN rate of 0.3143 indicated that nearly 31% of the customers who respond positively were misclassified as dissatisfied customers.
* Furthermore a low specificity of 0.3367 showed that the model fails to detect dissatisfied customers in the campaign, leading to the loss of potential customers of the business.

Q5. (b)

(i) (4 marks)

$$isclassification = 1 - \frac{1350 + 6500}{10000} = 0.215$$

$$sensitivity = \frac{6500}{500 + 6500} = 0.9286$$

$$specificity = \frac{1350}{1650 + 1350} = 0.45$$

$$F_1 = \frac{2 * 6500}{(2 * 6500) + 500 + 1650} = 0.8581$$

(ii) (3 marks)

$Cost_{TP} = -40$, $Cost_{FP} = 10$

$$cost\ per\ record = \frac{Cost_{TN} + Cost_{FP} + Cost_{FN} + Cost_{TP}}{N}$$
$$= \frac{0 + 10 * 1650 + 0 + (-40) * 6500}{10000} = -24.35$$

(iii) (7 marks)

Yes, because

- With unequal cost for TP and FP, the accuracy of model exceeding baseline performance, indicating that this model is fit for the data.
- The sensitivity and specificity of the unequal cost model is higher than sensitivity and specificity of the equal cost model, indicating that it is fitter than the equal cost model.
- High sensitivity in the unequal cost model indicated that the model able to identify customers that response positively to the campaign. This is supported by high F1 score.
- However, specificity of the model is still not meeting the threshold of 0.5, indicating most of the negative response records were misclassified as positive.
- In terms of cost per record, a great increase of 24.35 folds in model fitness for the unequal cost model when compared to the cost per record in the equal cost model, which is 17.21.
- The cost per record can further transformed into RM24.35 profit gained per record.

$$\text{cost per record (with equal cost)} = \frac{10 * 1990 + (-40) * 4800}{10000} = -17.21$$

[Total : 25 marks]